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Multimodal models for COVID-19 mortality prediction

INF-GRAM Practical 1, critical reading of an ML study

Wu JT, Armengol de la Hoz MÁ, Kuo PC, et al.
Journal of Digital Imaging 2022;35:1514–1529.

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1. Clinical question: who dies within 30 days of admission?

Population

Hospitalised adults (over 16) with COVID-19 in Madrid, Hoboken (USA) and Seoul

Task

Prognosis: binary risk prediction from EHR data and chest X-ray (CXR)

Target

Death within 30 days of hospital admission

Point of use

Admission: first CXR, first ED vitals, labs within 24 h → triage, level of care

Watch-outs: labs up to 24h make it a “day-1” model · early Madrid cases were diagnosed partly *from the CXR* · Seoul = “ICU” patients with 1.6% mortality

Wu et al., J Digit Imaging 2022, pp. 1515–1519

2. Data: three cohorts, three countries

Madrid	Hoboken	Seoul
1,628 included / 2,547 screened Development (4-fold CV)	201 included / 242 screened External test	315 included / 336 screened External test
189 deaths at 30 days (11.6%)	**87 deaths at 30 days (43.3%)**	**5 deaths at 30 days (1.6%)**
Dec 2019 – Jun 2020 17-hospital private network	Mar – Apr 2020 Community hospital	Jan – Dec 2020 Academic hospital (ICU)

Predictors: age, sex · 12 comorbidities · first ED vitals · ~25 labs (LDH, CRP, D-dimer, blood counts) · first frontal CXR
Reference standard: death within 30 days, from each hospital's EHR

Tables 1–2, pp. 1516–1518

2. Data: missing data and representativeness

Outcome capture

Deaths come from hospital EHRs, so deaths after discharge are likely missed. Hoboken data were pulled on 21 April for admissions up to 11 April: some patients had under 30 days of follow-up

Missing data

820 Madrid patients (32%) excluded for no admission CXR. Labs over 50% missing dropped; median or mode imputation; variables absent at external sites set to 0

Representativeness

First wave only, private network. Hypertension 6% in Madrid vs 54% in Hoboken: comorbidity coding differs between sites, not only patients

Three cohorts are not one population: case mix, outcome capture and prevalence all differ.

Tables 1–2, case definitions and preprocessing, pp. 1516–1518

3. ML workflow: from Madrid folds to external sites

Madrid data

1,628 patients, 189 deaths, first wave

4-fold CV

Tuning (300+ runs) and internal results on the same folds

Refit

Best settings retrained on all of Madrid

External test

Hoboken and Seoul, run once by testers not involved in tuning

EHR model

Logistic regression, random forest, gradient boosting, XGBoost; randomized search. Final pick not stated in main text

CXR model

DenseNet-121 pre-trained on CheXpert / MIMIC-CXR, partly fine-tuned, with anatomical bounding-box augmentation

Late fusion

The CXR model's probability becomes one extra feature next to the EHR features; the same 4 algorithms are re-tuned

Fig. 1 and design-decision tables, pp. 1516, 1520–1522

3. ML workflow: four leakage risks

1. Tuning = reporting

The same 4 folds pick the best of 300+ runs and report internal results. No nested CV, no held-out internal test: optimistic

2. Preprocessing on all of Madrid

Robust scaler fitted “across the whole Madrid dataset” (p. 1518), imputation probably too, before the folds

3. Possible stacking leakage

Fusion is trained on CXR probabilities. Out-of-fold or in-sample? Not stated. In-sample would be over-confident

4. Threshold choice unclear

Cut-off minimises |sens − spec|. Fixed on Madrid, or re-chosen at each external site? Not stated.

What stays clean: *external tests by a separate team, no tuning on Hoboken or Seoul data*

Preprocessing and model-picking sections, pp. 1518–1522

4. Evaluation: metrics reported vs metrics needed

Reported

- AUROC, sensitivity, specificity, PPV, NPV, F1, accuracy
- 95% CIs from 1,000 bootstrap samples
- Results by sex (supplement)
- Geographic external validation, 2 countries

Missing

- **Calibration**: none, though prevalence ranges 1.6–43%
- AUPRC and decision-curve analysis
- Formal test of fusion vs EHR (e.g. DeLong)
- Comparison with clinical scores (SEIMC, 4C)
- Temporal validation beyond wave 1

Misleading here: Seoul accuracy 0.95 is below the 98.4% of predicting “everyone survives” · F1 across sites mostly tracks prevalence

Statistical analysis, p. 1518; Discussion, p. 1526

4. Main results across sites

Madrid (internal CV)	Hoboken	Seoul
AUROC fusion 0.85 (0.83-0.87) EHR 0.82 - CXR 0.81	AUROC fusion 0.76 (0.70-0.82) EHR 0.74 - CXR 0.72	AUROC fusion 0.95 (0.92-0.98) EHR 0.92 - CXR 0.90
Fusion: PPV 0.27 - F1 0.40	Fusion: PPV 0.71 - F1 0.69	Fusion: PPV 0.13 - F1 0.21

5 deaths in Seoul

Seoul’s 0.95 AUROC rests on five events (sensitivity CI 0.20-0.86). Fusion adds ≈ 0.02-0.03 AUROC over EHR alone; CIs overlap and no test is reported.

Tables 3–4 and Fig. 3, pp. 1523–1524

5. Interpretation: one strength, one key limitation

STRENGTH

Honest external validation

Independent testers, no tuning on external data, three countries, open code and documented design decisions (CLAIM checklist)

Better practice than most COVID-19 models reviewed by Wynants et al. and Roberts et al

KEY LIMITATION

The evidence can’t carry the claims

Seoul has 5 deaths; settings, outcome capture and prevalence differ. calibration is never checked

So “fusion is better everywhere” and generalizability are not demonstrated

Wynants et al., BMJ 2020 · Roberts et al., Nat Mach Intell 2021 (refs 9, 16 of the paper)

5. Before clinical deployment

Temporal validation

Trained before dexamethasone, vaccines and new variants

Recalibrate, then prospective test

Local recalibration at each site before any use

Beat simple scores

Show added value over SEIMC or 4C Mortality

Clinical utility

Decision curves tied to a real action (ICU bed, escalation)

Robustness and fairness

Missing inputs at inference; subgroups beyond sex

Regulation

EU MDR medical device; high-risk under the AI Act

EXPLAINABILITY CHECK

SHAP splits credit across correlated labs (RBC, Hct, Hb, MCH). Grad-CAM “confirms” a lung focus the model was trained to have, averaged over confident deaths only

Explainability, Figs. 4–6, pp. 1519, 1524–1526

Take-home

Good process, thin evidence:

a well-documented pipeline whose external results can't yet justify clinical use.

Thank you, questions?